

Predicting Next-Day Closing Prices for Cryptocurrencies

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What did I learn?

Data Preparation & Feature Engineering

Cleaning the Data: Learned how to clean and prepare the raw crypto-markets.csv file.

Choosing Useful Columns: Picked the important columns and removed the ones we didn't need.

Setting the Target: Chose what we wanted to predict the next day's closing price.

Creating Features: Created helpful features from the data.

Modeling & Evaluation

Two Models Used: Built and tested two different models:

- Linear Regression (simple model)
- Random Forest Regressor (more advanced)

Time-Based Data Split: Learned how to split the data into training and testing sets while keeping the time order, which is very important for time-series data.

Scaling Features: Used StandardScaler to make sure large numbers didn't affect the model too much.

Understanding the Results: Used different measurements to check how well the models worked: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R-squared (R^2).

Comparing Models: Saw how each model handled the data. Random Forest worked better for complex patterns.

Seeing the Predictions: Plotted the predictions to see how close they were to the real prices.



What challenges did I face? How did I overcome those challenges?

1

Navigating Data Quality & Market Volatility

Missing/Inconsistent Data: crypto-markets.csv had missing or inconsistent historical data.

Strategically dropped rows with missing values resulting from feature engineering and carefully handled any infinite values.

2

Addressing Feature Limitations

Exclusion of External Factors: Significant challenge was the dataset's inherent lack of external factors (news, regulations, market sentiment).

My detailed feature engineering, incorporating a variety of technical indicators, was designed to capture as much of the underlying market dynamics as possible.

3

Predicting in a Dynamic Environment

Rapid Market Changes and Limited Accuracy: Cryptocurrency market behavior can change rapidly. This dynamic nature limits the overall accuracy of even sophisticated models in short-term forecasting.

Implemented both a simpler Linear Regression and a more complex Random Forest model to explore diverse approaches.

Ensured data split for training and testing preserved chronological order, making model evaluation more reflective of real-world prediction scenarios.

If I had more time, what would I add?

01

Incorporating External Factors

Top Priority: Integrate external data sources like news sentiment, social media trends, macroeconomic indicators, and government policy announcements.

02

Exploring Advanced Time-Series Models

Deep Learning: Delve into more advanced time-series specific models:

- Long Short-Term Memory (LSTM) networks
- Gated Recurrent Units (GRUs)

Sequential Data: These architectures are particularly well-suited for sequential data and capturing complex long-term dependencies in market trends.

What strengths do I bring to a team environment?

01

Strong Analytical & Problem-Solving Skills

Complex Problem Breakdown: From identifying relevant features to troubleshooting model performance, I demonstrated an ability to break down complex problems.

Thorough Analysis: Analyzed data thoroughly and devised effective solutions.

03

Adaptability & Continuous Learning

New Dataset & Techniques: Quickly adapted to working with a new, volatile dataset and successfully applied and compared different machine learning techniques.

Eagerness to Learn: Showcases my eagerness and capacity to rapidly learn and implement new concepts and tools efficiently.

